CardioTwin

Personalized Cardiovascular Digital Twin for Proactive Healthcare

Understand the present. Anticipate the next. Explore what could change.

Digital Twin Challenge 2026
Synthetic data only — not clinically validated.

From clinical snapshots to an evolving patient state

A generic threshold can miss context: patients have different baselines, activity patterns, histories, and recent trends.

CardioTwin combines:

- synthetic EHR history,
- simulated wearable-style physiology,
- trailing trends and personal baselines,
- calibrated 60-minute event forecasts,
- data-quality indicators and explanations,
- isolated what-if scenarios.

Research question: does combining static EHR, dynamic wearable features, and personal-baseline features improve a simulator-defined deterioration forecast on held-out synthetic patients?

More than a classifier: the digital twin

Each synthetic patient twin maintains:

Profile → Replay clock → Current state → Forecast → Scenario copy

Persistent state includes profile, observation history, baseline, rolling features, latest model prediction, data quality, SHAP explanation, and forecast/scenario history.

The deterministic replay clock advances one prerecorded five-minute observation at a time and never exposes future measurements to the current state.

Reproducible synthetic data

Current pilot:

- 60 synthetic patients
- 34,560 five-minute observations
- 90 rule-derived events
- 2 simulated days per patient

Full generator target: 500 patients × 7 days ≈ 1.0M observations.

The simulator produces HR, HRV, systolic/diastolic BP, steps, activity, sleep, sensor quality, missingness, noise, circadian variation, and injected deterioration trajectories.

Five-minute BP is an explicit simulation assumption, not a claim about typical consumer wearables.

A narrow simulator-defined event

Personal reference: median valid HR and systolic BP during the first 12 simulated hours.

A synthetic deterioration event is a contiguous run of at least **20 minutes** where:

- HR ≥ personal reference + **22 bpm**
- systolic BP ≥ personal reference + **17 mmHg**

The target is 1 for non-event observations during the **60 minutes before onset**.

Event logs and hidden simulator flags are excluded from model features. This endpoint is not clinically validated.

Integrated architecture

Synthetic EHR + wearable simulator + event module
→ leakage-safe feature engineering
→ patient-level train / validation / test
→ gradient-boosting ablations
→ validation-only Platt calibration and alert threshold
→ saved model
→ deterministic twin replay + what-if scenarios
→ FastAPI + SQLite audit storage
→ React / Recharts clinical research dashboard

SHAP TreeExplainer operates on the underlying gradient-boosting model for the current forecast.

Leakage-safe ML pipeline

1. Split patients approximately 70 / 15 / 15.
2. Build features from current and previous observations only.
3. Train logistic physiological baseline.
4. Train four gradient-boosting ablations.
5. Calibrate probabilities on validation patients with Platt scaling.
6. Choose prototype alert threshold on validation F1.
7. Evaluate once on held-out test patients.

No backward-fill, no event-log features, and no test-patient calibration.

Ablation study: personalization adds the clearest signal

Configuration	F1	ROC-AUC	PR-AUC
EHR only	0.000	0.521	0.036
Wearable only	0.232	0.802	0.153
EHR + wearable fusion	0.198	0.799	0.153
Fusion + personal baseline	**0.362**	**0.833**	**0.339**

The hypothesis is supported in this synthetic configuration, but these scores are not clinical evidence.

Genuine held-out synthetic metrics

For the calibrated personalized model:

- **ROC-AUC:** 0.833
- **PR-AUC:** 0.339
- **F1:** 0.362
- **Precision:** 0.587
- **Recall:** 0.262
- **Brier score:** 0.026
- **Validation-selected threshold:** 0.25

Event-level replay metrics:

- 100% event detection across 14 generated test events
- 26.1 simulated minutes mean lead time among detected events
- 1.72 false alerts per simulated patient-day

Robustness ROC-AUC: 0.822 with added sensor noise; 0.833 with simulated 15% current-reading loss.

Dashboard: one screen, real API state

The dashboard retrieves actual FastAPI responses and displays:

- patient selector and synthetic profile,
- latest HR / BP / personal baseline,
- recent physiological trajectory,
- calibrated 60-minute event probability,
- prototype alert category and data quality,
- SHAP contributing features,
- deterministic +5-minute replay controls,
- what-if scenario comparison,
- genuine held-out evaluation and ablations.

No successful API call or model update is faked in the UI.

What-if simulation without mutating the twin

1. Copy the current state.
2. Apply a declared hypothetical input rule.
3. Recalculate relevant features.
4. Run the same calibrated inference pipeline.
5. Compare baseline and hypothetical probabilities.
6. Preserve the original twin and log the scenario.

Included presets:

- return toward baseline,
- increased activity.

The result is **not** a causal treatment-effect estimate and does not recommend therapy or medication changes.

Responsible-use boundaries

- Synthetic patients and simulated measurements only.
- Forecasts a simulator-defined pattern, not heart attacks, cardiac arrest, or diagnoses.
- Risk categories are prototype presentation labels, not triage instructions.
- SHAP contributions are model explanations, not causal medical evidence.
- Generator shortcuts and event prevalence can make synthetic prediction easier than real-world prediction.
- No privacy/compliance claim has been made.
- Real-world clinical safety, fairness, prospective validity, workflow impact, and calibration remain untested.

Scale carefully. Validate honestly.

Next steps:

- generate the 500-patient / seven-day dataset,
- repeat experiments across multiple seeds,
- stress-test missingness, dropout, and baseline diversity,
- improve state persistence and replay history,
- compare additional calibrated classical models before deep learning,
- obtain clinical/domain review before any real-patient research.

CardioTwin already closes the technical demo loop: synthetic patient → replay → evolving twin → real model forecast → SHAP explanation → hypothetical scenario → held-out evaluation.

Synthetic research prototype — not clinically validated.